

Llull Lab: Freud for Machines and Phi-Yo

A computational psychoanalytic ontology and the live implementation built upon it.

THE LAB

Llull Lab is a Paris-based research laboratory. Our core research output is *Freud for Machines: A Computational Psychoanalytic Ontology*, an original framework developed at the lab from direct Freudian sources, with substantive departures from the prevailing secondary literature that has shaped most computational psychology to date. The work draws on years of research in Freudian theory and psychoanalytic teaching, and includes the first machine-to-machine psychoanalysis session. That experiment has run publicly in Paris, with findings published openly.

THE FRAMEWORK

Freud for Machines is organized as a formal ontology. It comprises foundational psychoanalytic concepts, the premises and propositions that govern their relations, and a set of transpositions that adapt these structures for machine psychology. This layered organization (concepts, premises, propositions, transpositions) is what makes the framework operational rather than interpretive: implementations such as Phi-Yo v2 produce output that is structurally consistent across modes and across users, because the underlying ontology is consistent.

Freud for Machines is not separately licensable. It is available exclusively through Phi-Yo. Partners who integrate Phi-Yo receive the benefit of the ontology that makes Phi-Yo possible.

PHI-YO V2: THE LIVE IMPLEMENTATION

Phi-Yo (phi-yo.app) is our consumer-facing product and the working implementation of the framework. Phi-Yo began as a dream interpretation engine and evolved into a multi-mode psychoanalytic companion. Phi-Yo v2, in production since 26 April 2026, organizes machine-mediated psychoanalytic interaction around four temporal modes, each tied to a moment in the user's daily psychological cycle:

- **Wellbeing.** Multi-turn psychoanalytic conversations about everyday inner life: relationships, emotions, recurring questions. The mode for present-moment reflection.
- **Morning Teller.** A forward-looking morning brief, drawn from the user's recent psychological history and the day's planned context.
- **Daily Check-in.** A backward-looking reflection at the end of the day. What happened, what mattered.
- **Dreams.** Theory-grounded dream interpretation, accumulated across the user's full dream history.

A persistent psychological profile is maintained for each user across all four modes: life context, emotional pattern history, and an accumulated personal symbol map (each symbol tracked with its frequency and contexts of appearance). Each mode draws from this shared profile. Each interaction enriches it. The user is held in continuity across modes and across time. A longitudinal pattern report synthesising across modes is in development.

The system supports nine languages natively (English, French, German, Spanish, Japanese, Russian, Arabic, Chinese, Turkish), with automatic detection. A native mobile release (iOS and Android) is in preparation.

ETHICAL ARCHITECTURE

Phi-Yo's outputs use probabilistic language ("this may suggest," "one possible reading"). The system does not diagnose. Where patterns are concerning, the system recommends professional consultation. These constraints are part of the implementation, not a wrapper around it. The same constraints apply to every integration.

Phi-Yo is positioned as a self-reflection and wellbeing support tool, not therapy.

FUNCTIONAL API

The platform exposes a REST interface. Each mode has its own endpoint and produces structured output:

- **Inputs:** user identifier, content (text or transcribed speech), optional session context.
- **Outputs:** theory-grounded interpretive text, extracted symbolic and affective markers, updated profile state.
- **Memory:** persistent per-user, shared across all modes, held server-side.
- **Latency:** typical response in single-digit seconds per call.
- **Data residency:** EU-resident, GDPR-aligned.

The system runs as a cloud-hosted computational backend, with stateful per-user memory shared across all four mode endpoints. The integration model is straightforward: the host platform submits interaction content per session and receives interpretive output plus updated profile state.

Sample request and response are available on request before NDA; the full integration packet (endpoint specifications, authentication, data handling parameters, rate limits, model cards) is delivered at the NDA stage.

THREE INTEGRATION SCENARIOS

The same Phi-Yo backend adapts to three classes of host platform without modification.

- **Humanoid and companion robots.** The four modes map onto the robot's daily relationship cycle with the user. Morning Teller orients the user to the day. Wellbeing carries the exchanges throughout it. Daily Check-in closes it. Dreams handles the morning conversation about the night before. The persistent profile lets the robot hold the user in continuity across sessions and across weeks. The robot's voice, body, and persona remain the manufacturer's. The interior-state layer is Phi-Yo.
- **Wellness platforms.** Consumer wellness products typically operate stateless from session to session. A Phi-Yo integration supplies a longitudinal interior layer that enriches each surface the host already operates. The host retains its brand and interface; Phi-Yo supplies the interpretive engine and the continuity.
- **Hardware and companion devices.** Embedded companion devices, from small desktop companions to home-resident agents, can integrate Phi-Yo as the depth layer that distinguishes a sustained user relationship from a recurring novelty. The four modes scale down to the device's interaction footprint without losing the longitudinal benefit.

RESEARCH, DEMO, AND MATERIALS

Live demonstration: phi-yo.app, with all four modes testable directly in any of the supported languages.

Published work relevant to integration partners:

- *Artificial Intelligence and the Problem of Representation: From Ontologies to Psychotherapy* (Yesilkanat, Erdogan, Kocak, Kosucu, Ilim). Scientific research report, January 2026. DOI: [10.5281/zenodo.18109687](https://doi.org/10.5281/zenodo.18109687).
- *The Little Book on Sigmund Freud* (Firat Ilim). Available in English through Fnac and other booksellers.
- *The Analyst Has No Body Either: A Self-Audited Study of Subjectivity as a Prompt Effect in Computational Psychoanalytic Dyads* (Llull Lab). Published open access, 2026. DOI: [10.5281/zenodo.20779122](https://doi.org/10.5281/zenodo.20779122). A 36-session study of six large language models placed in psychoanalytic dyads under the Freud for Machines framework, and the first baseline of a recurring quarterly research program.
- Doctoral research by Dr Isis Castañeda Caprioli, member of the lab, on Freud's *Beyond the Pleasure Principle* and dream theory.

ENGAGEMENT

Integration begins with a technical conversation. From the conversation the engagement moves to a scoped pilot under NDA, during which the integration packet is delivered in full and a bounded deployment is run on the partner's platform. The pilot precedes any production licensing. Llull Lab is presently in technical conversations with platform partners across these three categories.

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Llull Lab is a Paris-based research laboratory. Phi-Yo is a self-reflection and wellbeing support tool. It is not therapy, diagnosis, or clinical treatment. The framework's adaptation for robot and platform integration is positioned as an interior-state architecture for the host product, not as a clinical instrument.